

Documentation interfaces

This document contains the description of the interfaces used in the file “main_interfaz.f90”.

Note: the variables that contain an asterisk are **optional** arguments.

read_chemistry

- **Description:** Subroutine that reads all the chemical information (species, reactions, initial and boundary concentrations, etc).
- **Class: Chemistry**
- **Input:**

Variable	Type	Description
root	String	Root of input and output files of problem to be solved
path_pb	String	Path to input and output files of problem to be solved
path_DB	String	Path to chemical databases

- **Pre-process:**
 - Sets arbitrary input and output file units
- **Process:**
 - Reads **chemical data** written with CHEPROO-based format

write_conc_comp_wat_types

- **Description:** Subroutine that writes concentrations of aqueous components of initial and external water types in a file.
- **Class: Chemistry**
- **Input:**

Variable	Type	Description
path	String	Path to the output file
filename	String	Output file name

- **Process:**
 - Opens filename
 - Writes name & concentrations of aqueous components for each water type
 - Closes filename

get_num_aq_comps_dom

- **Description:** Function that gets the number of aqueous components in the domain. It assumes that all target waters in the domain have the same number of aqueous components.
- **Class: Chemistry**
- **Output:**

Variable	Type	Description
num_aq_comps	Integer	Number of aqueous components in the domain

- **Process:**
 - Gets the number of aqueous components of the first domain target water

interfaz_comps_arch

- **Description:** Subroutine that solves a reactive mixing iteration with **components**, using **files** as arguments. The integration method is **Euler explicit** and the reaction mixing ratios are **lumped**. It assumes that there are equilibrium reactions.

- **Class: Chemistry**

- **Input:**

Variable	Type	Description
path	String	Path to input and output files.
num_aq_comps	Integer	Nº aqueous components.
file_in	String	File containing concentrations of aqueous components after solving conservative transport iteration ($\tilde{\mathbf{u}}^k$).
Delta_t	Real	Time step (Δt^k)
file_out	String	File where concentrations of aqueous components after solving reactive mixing iteration will be written ($\mathbf{u}^{k+1}$).

- **Pre-process:**

- Allocate aqueous component concentrations after conservative transport ($\tilde{\mathbf{u}}^k$)
- Allocate chemical reactions contribution to aqueous component concentrations ($\mathbf{u}_K^k$)

- **Process:**

- **Read $\tilde{\mathbf{u}}^k$** from file_in. **Note:** The rows must represent components and the columns targets.
- **Solve** reactive mixing iteration for each target water:
 - **Loop over number of target waters in domain**
 - Compute chemical reactions contribution to aqueous component concentrations using **Euler explicit** and **lumping**:

$$\mathbf{u}_{K,j}^k = \Delta t^k \mathbf{U} \mathbf{S}_{K,nc}^T \mathbf{r}_{K,j}^k$$
 - Update the total aqueous component concentration in target water: $\mathbf{u}_j^{k+1} = \tilde{\mathbf{u}}_j^k + \mathbf{u}_{K,j}^k$
 - Speciate to compute variable activity species concentrations $\mathbf{c}_{nc,j}^{k+1}$ from components

- **Post-process:**

- **Write $\mathbf{u}^{k+1}$** in file_out (same dimensions as $\tilde{\mathbf{u}}^k$)
- **Write $\mathbf{c}_{nc}^{k+1}$** in file_out (rows: species, columns: targets)
- **Deallocate** arrays

interfaz_esp_arch

- **Description:** Subroutine that solves a reactive mixing iteration with **aqueous species**, using **files** as arguments. The integration method is **Euler explicit** and the reaction mixing ratios are **lumped**. It assumes that there are no equilibrium reactions.
- **Class: Chemistry**
- **Input:**

Variable	Type	Description
path	String	Path to input and output files.
num_aq_comps	Integer	Nº aqueous species
file_in	String	File containing concentrations of aqueous species after solving conservative transport iteration ($\tilde{\mathbf{c}}^k$).
Delta_t	Real	Time step (Δt^k)
file_out	String	File where concentrations of aqueous species after solving reactive mixing iteration will be written ($\mathbf{c}^{k+1}$).

- **Pre-process:**
 - Allocate aqueous species concentrations after conservative transport ($\tilde{\mathbf{c}}^k$)
 - Allocate chemical reactions contribution to aqueous species concentrations ($\mathbf{c}_K^k$)
- **Process:**
 - **Read $\tilde{\mathbf{c}}^k$ from file_in.** **Note:** The rows must represent species and the columns targets.
 - **Solve** reactive mixing iteration for each target water:
 - **Loop over number of target waters in domain**
 - Compute chemical reactions contribution to aqueous species concentrations using **Euler explicit** and **lumping**:
$$\mathbf{c}_{K,j}^k := \Delta t^k \mathbf{U} \mathbf{S}_{K,nc}^T \mathbf{r}_{K,j}^k$$
 - Update the total aqueous species concentrations in target water:
$$\mathbf{c}_j^{k+1} = \tilde{\mathbf{c}}_j^k + \mathbf{c}_{K,j}^k$$
- **Post-process:**
 - **Write $\mathbf{c}^{k+1}$ in file_out** (rows: species, columns: targets)
 - **Deallocate** arrays